

Reactive Transport Modelling in Porous Media using COMSOL-MATLAB-PHREEQC

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1. INTRODUCTION

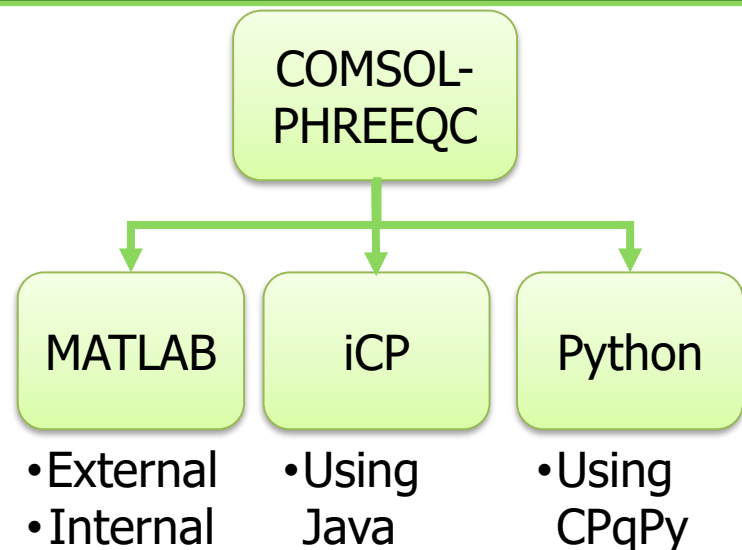


Fig 1. Coupling of COMSOL & PHREEQC.

2. OBJECTIVES

- ✓ To build a model to solve reactive transport in porous media
- ✓ To find a way to couple COMSOL– MATLAB – PHREEQC
- ✓ To validate the model with benchmark cases.

3. METHODOLOGY

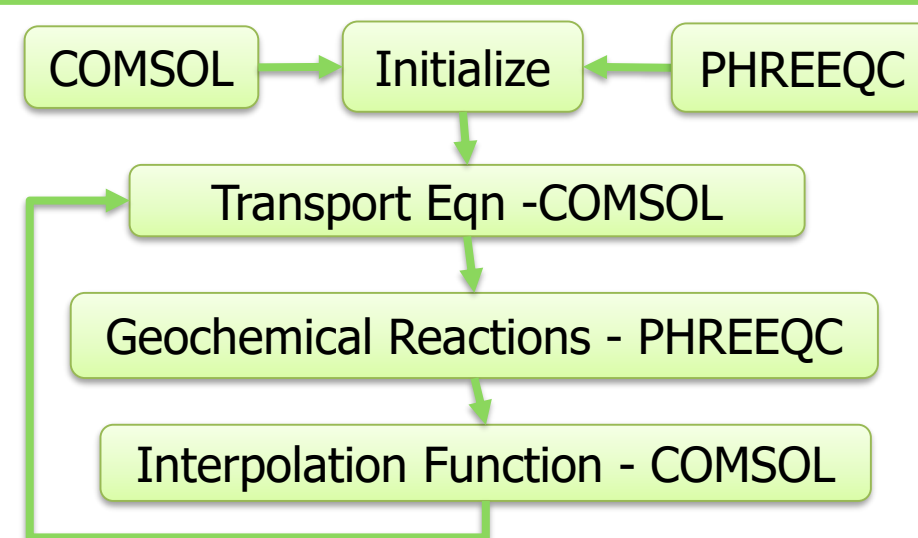


Fig 2. Methodology of the proposed model.

4. BENCHMARK VALIDATION 1 – PESTICIDE DEGRADATION

- ✓ This benchmark case is taken from Wissmeier(2011)^[1].
- ✓ This study analyzes the degradation kinetics of aldicarb and its toxic by-products.

Table 1. Model Parameters.

VARIABLES	UPPER LAYER	LOWER LAYER	UNIT
K_s	0.298	0.454	m/d
θ_s	0.399	0.339	
θ_r	0.001	0.001	
H_{p0}	0.01		m
$H_{p,init}$	$[-(z+1.2) - 0.2(z+0.4)]$	$[-(z+1.2)]$	m

- ✓ It focus on their decay timescales and spatial concentration distribution.

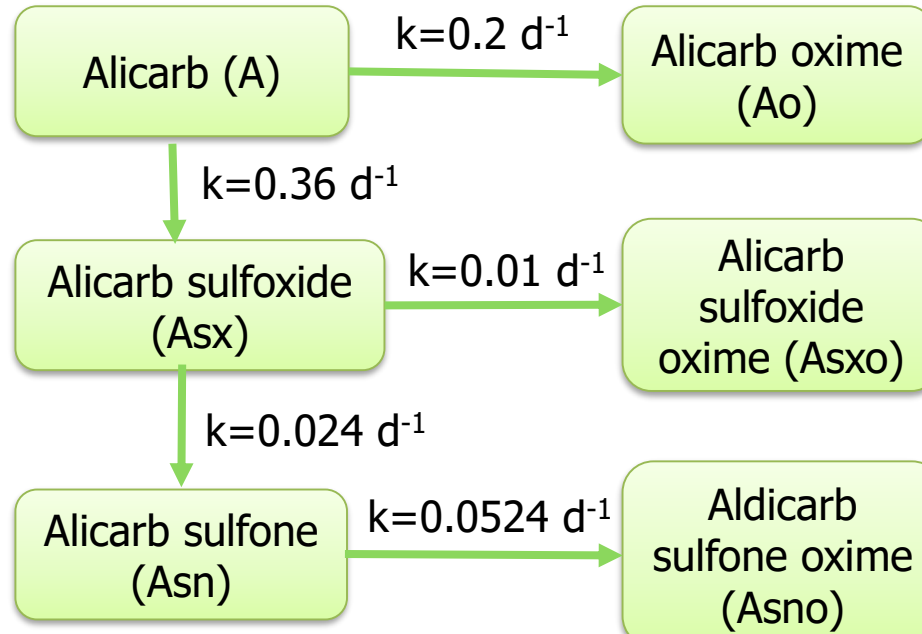


Fig 3. Reaction and rate constants for Aldicarb and its two products

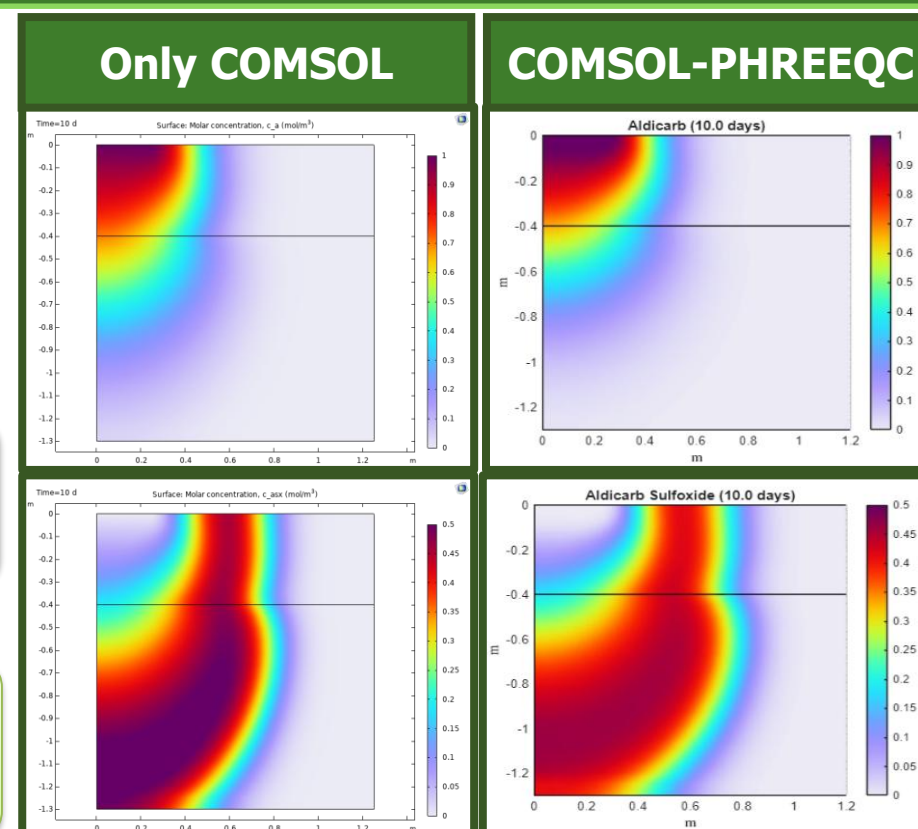


Fig 4. TOP – Molar Conc. of A (mol/m³). BOTTOM – Molar Conc. of Asx (mol/m³) after 10 days.

5. EXPERIMENTAL VALIDATION – CO₂ DISSOLUTION

- ✓ This experimental validation is taken from Wang (2024)^[2].
- ✓ Performed visual experiments of convective dissolution of high-pressure CO₂ into Ca(OH)₂.

$$\rho = \rho_0 \left[1 + \sum \beta_i C_i \right]$$

$$R = k_r C_{Ca} C_{CO_2}$$

Table 2. Parameters used in the model.

VARIABLES	VALUE
k	$1e^{-9} m^2$
ϕ	0.42
β_{CO_2}	$4.4e^{-6} m^3/mol$
β_{Ca}	$1.5e^{-4} m^3/mol$
C_{CO_2}	330 mol/m³
C_{Ca0}	10 mol/m³
k_r	10 m³/mol.s

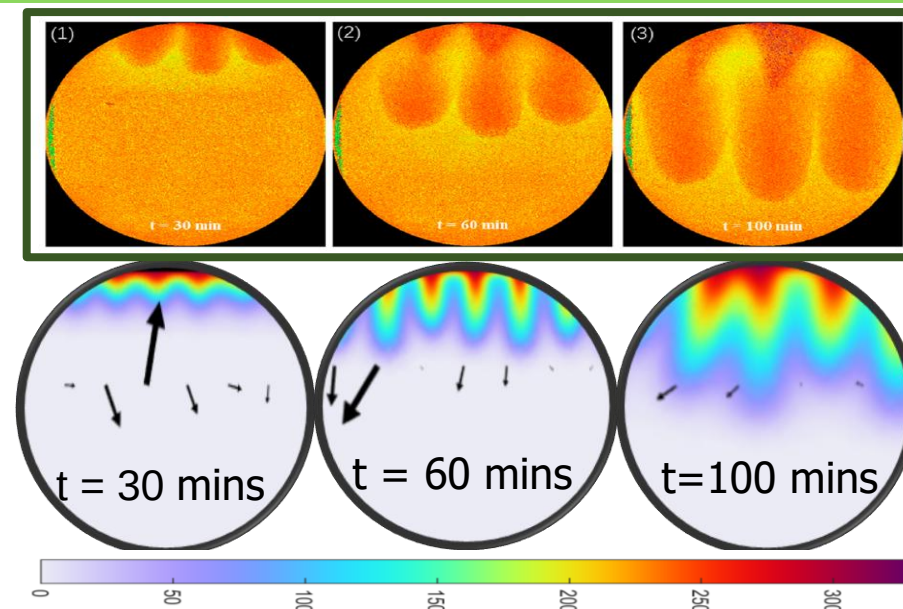


Fig 5. TOP – Experimental Results by Wang(2025)^[2]. BOTTOM - Using COMSOL and PHREEQC Coupling.

6. CONCLUSION

The benchmark study reveals that the coupled framework using COMSOL-MATLAB-PHREEQC can capture the evolution of contaminated plumes in soil and CO₂ dissolution.

7. FUTURE WORK

Implement this model to validate pore-scale model for multiphase flow & reactive transport in CO₂ storage.

8. REFERENCES

- Wissmeier, L., & Barry, D. A. (2011). Environmental Modelling & Software, 26(2), 210-218..
- Wang, L., et al. (2024). Advances in Water Resources, 190, 104746.